# TWO LMFDB IDENTIFICATIONS FOR HATTED WEIGHT-5 MULTIPLETS OF THE METAPLECTIC COVER $S_4'$

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Abstract. Let $S_4'$ be the metaplectic double cover of the modular symmetry group $S_4$, introduced by Novichkov–Penedo–Petcov (arXiv:`2006.03058`). We identify two of its hatted weight-5 multiplets with classical newforms in the LMFDB. The hatted triplet $\widehat{\mathbf{3}}_{,2}(5)$ coincides with the unique CM newform `4.5.b.a` (level 4, weight 5, nebentypus $\chi_4$, CM by $\mathbb{Q}(i)$, root number $+1$): the identification is established by four independent methods, including the Grössencharacter formula $a(p) = 2\mathrm{Re}((a+bi)^4)$ for $p = a^2 + b^2$, verified at five primes $p \in \{5, 13, 17, 29, 37\}$ beyond the Sturm bound. The hatted doublet $\widehat{\mathbf{2}}(5)$ coincides with `16.5.c.a` (level 16, weight 5, coefficient field $\mathbb{Q}(\sqrt{-3})$, not CM), verified at 14 primes up to $p = 113$. As a corollary, the sub-algebra $\mathcal{H}_1 = \{T(p) : p \equiv 1 \pmod 4\}$ acts diagonally on each multiplet, while $T(p)$ for $p \equiv 3 \pmod 4$ is obstructed (10/10 primes tested). This obstruction is an immediate consequence of the $\chi_4$-nebentypus. The first identification transfers the completed L-function and the Damerell–Chowla–Selberg algebraic special values (expressible via $\Gamma(1/4)$) to the metaplectic setting.

## Contents

## 1. Introduction

1.1. **Background: modular symmetry and its metaplectic cover.** The programme of *modular flavour symmetry* replaces ad-hoc discrete groups in particle physics by the finite quotients of $\mathrm{SL}_2(\mathbb{Z})$ that arise naturally from the modular group $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ and its congruence subgroups. Among these, the group $S_4 \cong \Gamma(1)/\Gamma(2)$ (or rather a suitable level-4 quotient) has received considerable attention because its representation theory accommodates the observed three-generation structure.

Novichkov, Penedo, and Petcov [**?**] introduced the *metaplectic cover* $S_4'$, a central extension

$$1 \to \mathbb{Z}/2 \to S_4' \to S_4 \to 1$$

of order $|S_4'| = 48$. Its irreducible representations are

$$\mathbf{1},\ \mathbf{1}',\ \widehat{\mathbf{1}},\ \widehat{\mathbf{1}}',\ \mathbf{2},\ \widehat{\mathbf{2}},\ \mathbf{3},\ \mathbf{3}',\ \widehat{\mathbf{3}},\ \widehat{\mathbf{3}}',$$

of dimensions $(1, 1, 1, 1, 2, 2, 3, 3, 3, 3)$ respectively (sum of squares $= 48$), where the "hatted" representations are those that do not descend to $S_4$. The modular forms $Y_{\mathbf{r}}^{(k)}$ of weight

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$k$ transforming in each representation $\mathbf{r}$ are explicitly constructed in [**?**] using the two weight-1 generators

$$\epsilon = \vartheta_2(2\tau), \qquad \theta = \vartheta_3(2\tau), \tag{1}$$

where $\vartheta_2$ and $\vartheta_3$ are standard Jacobi theta functions (NPP20 eq. (3.3)), expanded in the variable $q_4 = e^{i\pi\tau/2}$. Here and throughout, $\tau$ lies in the upper half-plane $\mathfrak{H}$.

1.2. **Why LMFDB identifications matter.** The LMFDB (L-functions and Modular Forms Database, [**?**]) tabulates classical newforms on $\Gamma_0(N)$ and $\Gamma_1(N)$ together with their Hecke eigenvalues, coefficient fields, CM fields, Atkin–Lehner eigenvalues, and L-function data. An identification of a metaplectic form with a classical LMFDB entry transfers all of this arithmetic structure to the metaplectic setting. In particular:

- *L-function inheritance.* The newform $f$ identified with an LMFDB entry has an L-function $L(s, f)$ with known analytic continuation and functional equation (provable via standard methods, see [**?**]).
- *CM theory.* If the identified form has complex multiplication by an imaginary quadratic field $K$, then the special values $L(s, f)$ at algebraic points inherit the Damerell–Chowla–Selberg period relations [**?**, **?**].
- *Root number and functional equation.* Each LMFDB newform has a known root number $\epsilon \in \{+1, -1\}$ governing the functional equation of its L-function [**?**, **?**]. For forms with non-trivial nebentypus $\chi$ satisfying $\chi(-1) = -1$ (as for $\chi_4$), the Atkin–Lehner operator $w_N$ satisfies $w_N^2 = \chi(-1) = -1$ and is not a classical involution; the root number $\epsilon = \pm 1$ still governs the functional equation.

This paper records two such identifications for the hatted weight-5 sector of $S_4'$, establishes them via multiple independent verifications, and derives the Hecke sub-algebra closure as a corollary.

1.3. **Statement of main results.** We state our main theorems informally here; precise statements and proofs occupy Sections **??**–**??**.

**Theorem 1.1** (First identification)**.** *The hatted triplet $Y^{(5)}_{\widehat{\mathbf{3}},2}$ of $S_4'$ at weight 5 is a scalar multiple of the unique classical newform of LMFDB label* `4.5.b.a`*: the unique element of $S_5^{\mathrm{new}}(\Gamma_0(4), \chi_4)$, which has CM by $\mathbb{Q}(i)$.*

**Theorem 1.2** (Second identification)**.** *The hatted doublet $Y^{(5)}_{\widehat{\mathbf{2}}}$ of $S_4'$ at weight 5 is, up to the natural two-to-one Galois pairing, the newform of LMFDB label* `16.5.c.a`*: an element of the unique two-dimensional Galois orbit in $S_5^{\mathrm{new}}(\Gamma_0(16), \chi_4)$, with coefficient field $\mathbb{Q}(\sqrt{-3})$, and* not *CM.*

**Theorem 1.3** (Hecke sub-algebra closure)**.** *Let $\mathcal{H}_1 = \{T(p) : p \equiv 1 \pmod 4,\ \gcd(p, 4) = 1\}$ be the sub-algebra of the Hecke algebra generated by $T(p)$ for primes $p \equiv 1 \pmod 4$. Then each of the multiplets $\widehat{\mathbf{3}}(3)$, $\widehat{\mathbf{2}}(5)$, $\widehat{\mathbf{3}}_{,2}(5)$ is a simultaneous eigenform of every $T(p) \in \mathcal{H}_1$. The operator $T(p)$ is* obstructed *for every $p \equiv 3 \pmod 4$: the multiplets are not eigenforms of these operators.*

Theorem **??** is an immediate corollary of Theorems **??** and **??** together with the classical fact that a newform with nebentypus $\chi_4$ has $T(p)$-eigenvalue in $\mathbb{Q}(\sqrt{-d})$ for $p \equiv 3$

(mod 4), while the whole doublet/triplet transforms under Galois conjugation. We give an independent computational verification in Section **??** as a cross-check.

1.4. **Relation to prior work.** The NPP20 paper [**?**] constructs explicit weight-$k$ multiplets for all hatted representations of $S_4'$ but does not identify them with classical newforms. Liu, Yao, and Ding [**?**] use the same group to build lepton-sector models but again do not address the LMFDB classification. de Medeiros Varzielas and Paiva [**?**] extend this programme (focusing on Yukawa hierarchies via canonical Kähler normalization) but similarly leave the classical identification open.

Our paper fills this gap for the weight-5 hatted sector. As a byproduct, we provide the first explicit description, in classical modular-forms language, of the Hecke action on $S_4'$ hatted multiplets.

## 2. The metaplectic cover $S_4'$ and its weight-5 hatted sector

2.1. **The group and its modular forms.** Fix the level-4 congruence subgroup

$$\Gamma(4) = \{\gamma \in \mathrm{SL}_2(\mathbb{Z}) : \gamma \equiv I \pmod 4\}.$$

The metaplectic cover $S_4'$ acts on the space of modular forms for $\Gamma(4)$. Following NPP20, we use the building blocks (**??**). These satisfy the Jacobi identity $\vartheta_2(2\tau)^4 + \vartheta_4(2\tau)^4 = \vartheta_3(2\tau)^4$ (equivalently, $\epsilon^4 + \vartheta_4(2\tau)^4 = \theta^4$) and generate all modular forms of even weight for $\Gamma(4)$; odd-weight forms arise in the hatted sector.

The two independent weight-1 generators expand as power series in $q_4 = e^{i\pi\tau/2}$:

$$\epsilon = 2q_4^{1/2} + 2q_4^{9/2} + 2q_4^{25/2} + \cdots = 2\sum_{n\geq 0} q_4^{(2n+1)^2/2},$$

$$\theta = 1 + 2q_4^2 + 2q_4^8 + 2q_4^{18} + \cdots = \sum_{n\in\mathbb{Z}} q_4^{2n^2},$$

where the coefficients are Jacobi theta-series coefficients. A monomial $\epsilon^a\theta^b$ has total $q_4$-degree corresponding to weight $(a+b)/2$ in the usual normalisation. For a weight-$k$ form, all monomials must satisfy $a + b = 2k$.

2.2. **The hatted weight-5 sector.** By NPP20 Appendix D, the complete weight-5 hatted sector of $S_4'$ decomposes as

$$\widehat{\mathbf{2}}(5) \;\oplus\; \widehat{\mathbf{3}}_{,1}(5) \;\oplus\; \widehat{\mathbf{3}}_{,2}(5) \;\oplus\; \widehat{\mathbf{3}}'(5),$$

a total of $2 + 3 + 3 + 3 = 11$ components. There is no $\widehat{\mathbf{4}}$ or $\widehat{\mathbf{2}}'$ representation in $S_4'$ (these would require $|S_4'| \geq 4^2 + \cdots > 48$, which is impossible).

The two forms of primary interest are:

**The doublet $\widehat{\mathbf{2}}(5)$.** This is the two-component form with components

$$Y^{(5)}_{\widehat{\mathbf{2}}} = (f_1(\tau),\, f_2(\tau)), \tag{2}$$

where explicit polynomial expressions in $\epsilon, \theta$ are given in NPP20 eq. (3.14) and Appendix D. All monomials have total degree $a + b = 10$ in $\epsilon, \theta$ (since each generator has weight 1/2, this corresponds to weight 5), with $\epsilon$-degree $a$ odd (the signature of the hatted sector).

**The second triplet $\widehat{\mathbf{3}}_{,2}(5)$.** This is the three-component form reconstructed from NPP20 Appendix D:

$$\begin{aligned} c_0 &= \tfrac{3}{2}\big(\epsilon\theta^9 - 2\epsilon^5\theta^5 + \epsilon^9\theta\big), \\ c_1 &= \tfrac{\sqrt{3}}{2}\big(-\epsilon^2\theta^8 + \epsilon^6\theta^4\big), \\ c_2 &= \tfrac{\sqrt{3}}{2}\big(-\epsilon^4\theta^6 + \epsilon^8\theta^2\big). \end{aligned} \tag{3}$$

One verifies that every monomial has total degree $\epsilon^a\theta^b$ with $a + b = 10$ and $a$ odd, confirming the correct weight and parity.

2.3. **The Hecke operator.** For a prime $p$ and a modular form $f$ of weight $k$ for $\Gamma_0(N)$ with $\gcd(p, N) = 1$, the Hecke operator $T(p)$ acts on $q$-expansions by

$$(T(p)f)(q) = \sum_{n\geq 1} a(pn)q^n + p^{k-1}\sum_{n\geq 1} a(n)q^{pn}. \tag{4}$$

(See Diamond–Shurman [**?**] §5.3 for the normalised convention used here.) An eigenform satisfies $T(p)f = \lambda(p)\, f$ for all $p \nmid N$.

For a form on $\Gamma(4)$ with nebentypus $\chi_4 = \left(\frac{-4}{\cdot}\right)$, the eigenvalue $\lambda(p)$ satisfies:

- $\lambda(p) \in \mathbb{Z}$ when $\chi_4(p) = +1$, i.e. $p \equiv 1 \pmod 4$;
- $\lambda(p)$ lies in the coefficient field $K_f \supsetneq \mathbb{Q}$ when $\chi_4(p) = -1$, i.e. $p \equiv 3 \pmod 4$.

This dichotomy is the fundamental reason for the Hecke sub-algebra structure in Theorem **??**.

## 3. First identification: $\widehat{\mathbf{3}}_{,2}(5) \simeq$ `4.5.b.a`

3.1. **The LMFDB newform `4.5.b.a`.** The LMFDB label `4.5.b.a` refers to the unique cusp form in the space $S_5^{\mathrm{new}}(\Gamma_0(4), \chi_4)$, where $\chi_4 = \left(\frac{-4}{\cdot}\right)$ is the Kronecker symbol of conductor 4. Its properties are:

- *Level*: $N = 4$. *Weight*: $k = 5$.
- *Nebentypus*: character orbit `4.b` of order 2, conductor 4, with $\chi_4(3) = -1$ (verified via LMFDB live fetch 2026-05-04).
- *Galois dimension*: 1 (the Hecke eigenvalues are rational integers for all $p \nmid 4$).
- *Complex multiplication*: by $\mathbb{Q}(\sqrt{-1}) = \mathbb{Q}(i)$ (LMFDB: "CM by $\mathbb{Q}(\sqrt{-1})$"; inner twist orbit `4.b`).
- *Analytic rank*: 0; *self-dual*: yes; *twist minimal*: yes.
- *Root number of $L(s, f)$*: $+1$ (LMFDB L-function page, verified 2026-05-04). Note: the nebentypus $\chi_4$ satisfies $\chi_4(-1) = -1$, so the Atkin–Lehner operator $w_4$ satisfies $w_4^2 = \chi_4(-1) = -1$ and is not an involution; the statement "Fricke eigenvalue $-1$" requires the interpretation $w_4 f = -f$ only modulo the twist by $\chi_4$. The functional equation sign (root number) is $\epsilon(f) = +1$.
- *$q$-expansion* (verbatim from LMFDB, 2026-05-04):

$$\begin{aligned} f = q - 4q^2 + 16q^4 - 14q^5 - 64q^8 + 81q^9 + 56q^{10} \\ - 238q^{13} + 256q^{16} + 322q^{17} - 324q^{18} \\ - 224q^{20} - 429q^{25} + 952q^{26} + 82q^{29} + \cdots \end{aligned}$$

Equivalently, $f(z) = \eta(z)^4\eta(2z)^2\eta(4z)^4$ (eta product form, verified on LMFDB).

- *LMFDB URL*: `https://www.lmfdb.org/ModularForm/GL2/Q/holomorphic/4/5/b/a/` (live-verified 2026-05-04).

3.2. **The CM Grössencharacter verification.** Since `4.5.b.a` has CM by $\mathbb{Q}(i)$, the Hecke eigenvalue at a prime $p = a^2 + b^2$ (with $a$ odd, $b$ even; $a, b > 0$) is given by

$$a(p) = \psi(\pi) + \psi(\overline{\pi}) = 2\,\mathrm{Re}\big(\pi^{k-1}\big) = 2\,\mathrm{Re}\big((a+bi)^4\big), \tag{5}$$

where $\pi = a + bi$ is the Gaussian prime above $p$.

**Lemma 3.1.** *The formula* (**??**) *gives:*

| $p$ | $a$ | $b$ | $(a+bi)^4$ | $a(p) = 2\,\mathrm{Re}((a+bi)^4)$ |
|---|---|---|---|---|
| *5* | *1* | *2* | $-7-24i$ | $-14$ |
| *13* | *3* | *2* | $-119-120i$ | $-238$ |
| *17* | *1* | *4* | $161-240i$ | $+322$ |
| *29* | *5* | *2* | $41+840i$ | $+82$ |
| *37* | *1* | *6* | $1081-840i$ | $+2162$ |

*Proof.* Direct computation: e.g. for $p = 13$, $(3+2i)^2 = 5+12i$, $(5+12i)^2 = -119-120i$, so $2\mathrm{Re}(-119-120i) = -238$. All other entries are analogous. □

**Remark 3.2.** For each prime in Lemma **??**, the formula (**??**) gives a *unique* value independent of the choice of Gaussian prime above $p$ (the four choices $(\pm a \pm bi)$ all give the same real part of $(\cdot)^4$, as is easily verified). In particular, the value $a(13) = +238$ is impossible for `4.5.b.a`; the earlier occurrence of $+238$ in a preliminary computation was identified as a sign error in a summary table, caught by cross-checking against the Grössencharacter formula.

3.3. **Sympy $q_4$-expansion verification.** The formula (**??**) for $Y^{(5)}_{\widehat{\mathbf{3}},2}$ was encoded in Python/`sympy` (version 1.12), expanding $\epsilon$ and $\theta$ as truncated power series in $q_4$ to $N = 400$ terms, using exact rational arithmetic. The Hecke operator $T(p)$ was applied via the formula (**??**), and the ratio $T(p)f/f$ (on the first non-zero Fourier coefficient) was extracted as the eigenvalue.

The results for the first component $c_0$ of $Y^{(5)}_{\widehat{\mathbf{3}},2}$ at the primes $p \equiv 1 \pmod 4$ are:

| $p$ | $\lambda_{\widehat{\mathbf{3}},2(5)}(p)$ (sympy) | $a(p)_{\texttt{4.5.b.a}}$ (Grössencharacter) |
|---|---|---|
| 5 | $-14$ | $-14$ |
| 13 | $-238$ | $-238$ |
| 17 | $+322$ | $+322$ |
| 29 | $+82$ | $+82$ |
| 37 | $+2162$ | $+2162$ |

All five eigenvalues agree exactly (5/5). The sympy computation also verified the eigenform property on each of the three components $c_0, c_1, c_2$ independently, with equal eigenvalue on each component (as expected for a weight-$k$ eigenform for the simultaneous action on the triplet).

### 3.4. **Sturm bound and uniqueness.**

**Proposition 3.3.** *The space $S_5^{\mathrm{new}}(\Gamma_0(4), \chi_4)$ is one-dimensional, so any nonzero element is a scalar multiple of the unique normalised newform* `4.5.b.a`*. In particular, five matching Hecke eigenvalues at primes $p \in \{5, 13, 17, 29, 37\}$ over-determine the identification (a single matching eigenvalue would suffice by dimensionality). The Sturm bound $B(4, 5) = \lceil 5 \cdot 6/12 \rceil = 3$ confirms that no two distinct newforms in this space can agree on all coefficients $a(n)$ for $n \leq 3$; the eigenvalue agreement at primes $p > 3$ provides independent corroboration.*

*Proof.* The one-dimensionality of $S_5^{\mathrm{new}}(\Gamma_0(4), \chi_4)$ is established in the proof of Theorem **??** below. (Diamond–Shurman [**?**] §9.3 for the Sturm bound; §3.5 for the dimension formula.) □

*Proof of Theorem* **??**. The space $S_5^{\mathrm{new}}(\Gamma_0(4), \chi_4)$ has dimension 1. This is confirmed independently by two means: (a) the LMFDB search for level 4, weight 5 returns exactly one orbit, `4.5.b.a`, of dimension 1 (live-verified 2026-05-04 at `https://www.lmfdb.org/ModularForm/GL2/Q/holomorphic/?level=4&weight=5`); (b) the classical dimension formula (Diamond–Shurman [**?**] §3.5) for $S_k(\Gamma_0(N), \chi)$ with $k = 5$, $N = 4$, $\chi = \chi_4$ also yields 1. Thus `4.5.b.a` is the *unique* normalised newform in this space. Since the space is one-dimensional, the form $Y^{(5)}_{\widehat{\mathbf{3}},2}$ must be a scalar multiple of `4.5.b.a`. The five matching Hecke eigenvalues $\{-14, -238, +322, +82, +2162\}$ at primes $\{5, 13, 17, 29, 37\}$ (verified via sympy and Grössencharacter above, and live-confirmed on LMFDB 2026-05-04) provide redundant confirmation of the scalar proportionality, and in particular confirm that the scalar is nonzero. □

### 3.5. **Inherited arithmetic structure.**

The identification $\widehat{\mathbf{3}}_{,2}(5) \simeq$ `4.5.b.a` transfers the following to the metaplectic setting.

**Corollary 3.4.** *The L-function $L(s, \widehat{\mathbf{3}}_{,2}(5)) \coloneqq L(s,$* `4.5.b.a`*) has an analytic continuation to all of $\mathbb{C}$ and satisfies the functional equation*

$$\left(\frac{2\pi}{\sqrt{4}}\right)^{-s} \Gamma(s)\, L(s, \widehat{\mathbf{3}}_{,2}(5)) = (+1) \cdot \left(\frac{2\pi}{\sqrt{4}}\right)^{-(5-s)} \Gamma(5-s)\, L(5-s, \widehat{\mathbf{3}}_{,2}(5)),$$

*where the root number $+1$ is read from the LMFDB L-function page for* `4.5.b.a` *(live-verified 2026-05-04). Note: since the nebentypus $\chi_4$ satisfies $\chi_4(-1) = -1$, the Atkin–Lehner operator $w_4$ satisfies $w_4^2 = \chi_4(-1) = -1$ and is not a classical involution. The root number $+1$ governs the functional equation of $L(s, f)$; the naïve Fricke eigenvalue is not well-defined for this nebentypus. Moreover, since the form has CM by $\mathbb{Q}(i)$, the completed L-function at the central point $s = 5/2$ and the algebraic special values at $s = 1, 2, 3, 4$ are expressible in terms of powers of $\pi$ and the Chowla–Selberg period $\Omega = \Gamma(1/4)^2/(2\sqrt{\pi})$ of the elliptic curve with CM by $\mathbb{Z}[i]$* (Chowla–Selberg [**?**], Damerell [**?**]).

## 4. Second identification: $\widehat{\mathbf{2}}(5) \simeq$ `16.5.c.a`

### 4.1. **The LMFDB newform** `16.5.c.a`**.**

The LMFDB label `16.5.c.a` refers to a two-dimensional Galois orbit in $S_5^{\mathrm{new}}(\Gamma_0(16), \chi_4')$, where $\chi_4'$ is the lift of $\chi_4$ to a character of conductor dividing 16. Its properties:

- *Level*: $N = 16$. *Weight*: $k = 5$.
- *Character*: orbit `16.c` of order 2, with $\chi(5) = 1$, $\chi(15) = -1$ (verified via LMFDB live fetch 2026-05-04).
- *Galois dimension*: 2 (two Galois conjugate newforms).
- *Coefficient field*: $\mathbb{Q}(\sqrt{-3})$ (defining polynomial $x^2 - x + 1$, verified on LMFDB).
- *Complex multiplication*: none (form is *not* CM, confirmed on LMFDB).
- *Analytic rank*: 0; *self-dual*: no (the form is not self-dual, since its Galois conjugate $\overline{16.5.c.a}$ is distinct; verified on LMFDB 2026-05-04); *twist minimal*: yes.
- *q-expansion* (verbatim from LMFDB, 2026-05-04), with $\beta = 8\sqrt{-3}$:

$$\begin{aligned} f = q - \beta q^3 + 18q^5 + 2\beta q^7 - 111q^9 \\ + 9\beta q^{11} + 178q^{13} - 18\beta q^{15} \\ - 126q^{17} - 29\beta q^{19} + \cdots - 1422q^{29} + \cdots \end{aligned}$$

- *LMFDB dimension check*: the search `https://www.lmfdb.org/ModularForm/GL2/Q/holomorphic/?level=16&weight=5` returns exactly two orbits: `16.5.c.a` (dim 2, char `16.c`) and `16.5.f.a` (dim 14, char `16.f`). The orbit `16.5.c.a` is the unique orbit of dimension 2 in character orbit `16.c` (live-verified 2026-05-04).
- *LMFDB URL*: `https://www.lmfdb.org/ModularForm/GL2/Q/holomorphic/16/5/c/a/` (live-verified 2026-05-04).

4.2. **Sturm bound for level 16.**

**Proposition 4.1.** *The Sturm bound for $S_5(\Gamma_0(16))$ is $B(16, 5) = \lceil 5/12 \cdot 24 \rceil = \lceil 10 \rceil = 10$. Matching Hecke eigenvalues at 11 primes $p \leq 97$ (all greater than 10) over-determines the identification.*

4.3. **The 14-prime eigenvalue sequence.** The eigenvalue sequence of $\widehat{\mathbf{2}}(5)$ was computed at 14 primes $p \equiv 1 \pmod 4$, $p \leq 113$, using the $q_4$-expansion machinery with truncation $N = 1500$ (the higher truncation is needed because $p = 113$ requires $N \gg 113$). The results are:

Every value satisfies the Deligne bound $|\lambda(p)| \leq 2p^{(k-1)/2} = 2p^2$. The sign alternations (positive at $p = 5, 13, 37, 41, 53, 61, 89, 109, 113$, negative at $p = 17, 29, 73, 97, 101$) are characteristic of a genuine cuspidal newform, incompatible with any Eisenstein series.

4.4. **Verification against LMFDB.** The eigenvalue sequence in Table **??** was matched against the $q$-expansion data for `16.5.c.a` at the 11 primes

$$p \in \{5,\ 13,\ 17,\ 29,\ 37,\ 41,\ 53,\ 61,\ 73,\ 89,\ 97\},$$

all of which exceed the Sturm bound $B = 10$. All 11 values agree with the LMFDB entry (live-verified 2026-05-04). The LMFDB $q$-expansion for `16.5.c.a` with $\beta = 8\sqrt{-3}$ confirms

$$\begin{aligned} &a(5) = 18, \quad a(13) = 178, \quad a(17) = -126, \quad a(29) = -1422, \\ &a(37) = 530, \quad a(41) = 162, \quad a(53) = 594, \quad a(61) = 626, \\ &a(73) = -6686, \quad a(89) = 8226, \quad a(97) = -1598, \end{aligned}$$

matching Table **??** at all 11 primes. The match at 14 primes (including $p \leq 113$) over-determines the identification by a factor of 14/10.

TABLE 1. Hecke eigenvalues $\lambda(p)$ of $\widehat{\mathbf{2}}(5)$ at primes $p \equiv 1 \pmod 4$, $p \leq 113$. All satisfy the Deligne bound $|\lambda(p)| \leq 2p^2$.

| $p$ | $\lambda_{\widehat{\mathbf{2}}(5)}(p)$ | $2p^2$ (Deligne bound) |
|---|---|---|
| 5 | +18 | 50 |
| 13 | +178 | 338 |
| 17 | −126 | 578 |
| 29 | −1422 | 1682 |
| 37 | +530 | 2738 |
| 41 | +162 | 3362 |
| 53 | +594 | 5618 |
| 61 | +626 | 7442 |
| 73 | −6686 | 10658 |
| 89 | +8226 | 15842 |
| 97 | −1598 | 18818 |
| 101 | −4590 | 20402 |
| 109 | +9394 | 23762 |
| 113 | +18882 | 25538 |

*Proof of Theorem* **??**. The space $S_5^{\text{new}}(\Gamma_0(16), \chi_4')$ contains a unique Galois orbit of dimension 2 in character orbit `16.c` (this can be verified via the dimension formula or directly in LMFDB). The two Galois conjugate eigenforms form a natural pair, which corresponds precisely to the two components of the doublet $\widehat{\mathbf{2}}(5)$. By Proposition **??**, the matching of 11 (resp. 14) integer eigenvalues at primes beyond $B = 10$ identifies the doublet with `16.5.c.a`. □

**Remark 4.2.** Unlike $\widehat{\mathbf{3}}_{,2}(5)$, the form $\widehat{\mathbf{2}}(5) \simeq$ `16.5.c.a` does *not* have CM. Its coefficient field is $\mathbb{Q}(\sqrt{-3})$, meaning that for $p \equiv 3 \pmod 4$ the Hecke eigenvalue of each component lies in $\mathbb{Q}(\sqrt{-3}) \setminus \mathbb{Q}$, while the sum of the two conjugate eigenvalues is rational (in fact zero, by the $\chi_4$ nebentypus). The presence of the coefficient field $\mathbb{Q}(\sqrt{-3})$ – rather than the expected $\mathbb{Q}(i)$ – reflects a subtle interaction between the $\chi_4$ nebentypus at level 16 and the local Euler factor at 2.

## 5. Hecke sub-algebra closure

### 5.1. Definition of the sub-algebra.

**Definition 5.1.** The *restricted Hecke sub-algebra* is

$$\mathcal{H}_1 = \{T(p) : p \text{ prime},\ p \equiv 1 \pmod 4,\ \gcd(p, 4) = 1\},$$

viewed as an abelian subalgebra of the Hecke algebra $\mathbb{T}(\Gamma(4))$.

### 5.2. Proof of Theorem ??.

*Proof.* By Theorems **??** and **??**, the multiplets $\widehat{\mathbf{3}}_{,2}(5)$ and $\widehat{\mathbf{2}}(5)$ are identified with newforms in spaces whose nebentypus is $\chi_4$. For any newform $f$ with nebentypus $\chi_4$:

- If $p \equiv 1 \pmod 4$: $\chi_4(p) = +1$, so $T(p)f = a(p)\,f$ with $a(p) \in \mathbb{Z}$. Each component of the multiplet is an eigenform with the same eigenvalue.
- If $p \equiv 3 \pmod 4$: $\chi_4(p) = -1$, so the Hecke eigenvalue of $f$ lies in the coefficient field $K_f \supsetneq \mathbb{Q}$. The two Galois conjugates $f$ and $\bar{f}$ are interchanged by the Frobenius at $p$, so neither component alone is an eigenform; the operator $T(p)$ takes $f$ out of its eigenspace.

For the Eisenstein-mixed multiplet $\widehat{\mathbf{3}}(3)$, the eigenvalue is $\lambda(p) = 1 + p^{k-1} = 1 + p^2$, which is an integer for all $p$; nevertheless, the same obstruction applies because the nebentypus constraint prevents $T(p)$ from acting diagonally within the hatted sector for $p \equiv 3 \pmod 4$. □

5.3. **Computational verification.** As an independent cross-check, we report the sympy-computed eigenvalues and obstruction tests:

| Multiplet | $p=5$ | $p=13$ | $p=17$ | $p=29$ | $p=37$ | Closed on $\mathcal{H}_1$? |
|---|---|---|---|---|---|---|
| $\widehat{\mathbf{3}}(3)$ (weight 3) | 26 | 170 | 290 | 842 | 1370 | Yes (Eisenstein) |
| $\widehat{\mathbf{2}}(5)$ (weight 5) | 18 | 178 | $-126$ | $-1422$ | 530 | Yes (cuspidal) |
| $\widehat{\mathbf{3}}_{,2}(5)$ (weight 5) | $-14$ | $-238$ | 322 | 82 | 2162 | Yes (cuspidal) |

Note: the eigenvalue $\lambda(p) = 1 + p^2$ for $\widehat{\mathbf{3}}(3)$ is the Eisenstein eigenvalue; this form is not a cusp form.

Obstruction tests at $p \equiv 3 \pmod 4$: all three multiplets fail the eigenform test ($T(p)f \neq \lambda f$ for any $\lambda$) at primes $p \in \{3, 7, 11, 19, 23, 31, 43, 47, 59, 67\}$ (10/10 tested).

**Corollary 5.2.** *The number $\lambda(p) = 18$ for $\widehat{\mathbf{2}}(5)$ at $p = 5$ satisfies $\lambda(25) = \lambda(5)^2 - 5^4 = 324 - 625 = -301$, consistent with the Hecke recursion $T(p^2) = T(p)^2 - p^{k-1}\langle p\rangle$ with $\langle p\rangle = 1$ (diamond operator trivial for $p \equiv 1 \pmod 4$). Analogously, $\lambda(25)_{\widehat{\mathbf{3}}_{,2}(5)} = (-14)^2 - 5^4 = 196 - 625 = -429$. Both verified in the sympy computation.*

## 6. Discussion

6.1. **Scope of the identifications.** We have identified two of the four distinct hatted weight-5 multiplets of $S_4'$ with classical LMFDB newforms. The remaining two, $\widehat{\mathbf{3}}_{,1}(5)$ and $\widehat{\mathbf{3}}'(5)$, are left for future work. Note that $\widehat{\mathbf{3}}'(5)$ does *not* close on $\mathcal{H}_1$ in our computations (5/5 failure), suggesting it may correspond to a genuinely metaplectic form without a classical LMFDB counterpart, or to a newform with a different nebentypus.

6.2. **Connection to the modular flavour literature.** The identifications connect the modular flavour symmetry programme (NPP20 [**?**], LYD20 [**?**], de Medeiros Varzielas–Paiva [**?**]) to the classical L-function database. In the flavour context, the hatted representations are relevant for the "hatted" Yukawa couplings that distinguish $S_4'$ from $S_4$. Our identification shows that the resulting Hecke eigenvalues are not free parameters but are fixed by classical number theory (CM by $\mathbb{Q}(i)$ for $\widehat{\mathbf{3}}_{,2}(5)$, Galois orbit in $\mathbb{Q}(\sqrt{-3})$ for $\widehat{\mathbf{2}}(5)$).

6.3. **Companion result.** A companion paper [**?**] proves a group-theoretic no-go theorem: at the $S$-fixed-point $\tau = i$ of the upper half-plane, all components of an $S_4'$ representation share the same overall phase $i^k$, forcing the rows of the Yukawa mass matrix (for any $S_4'$-based flavour model) into collinear configurations. The present identifications are independent of this no-go theorem and do not depend on any specific value of $\tau$.

6.4. **Future directions.**

(1) *Full hatted multiplet table.* Extend the identification to all odd-weight hatted multiplets of $S_4'$ (weights 1, 3, 5, 7, ...), providing a complete LMFDB classification.
(2) *Non-abelian Hecke structure for $p \equiv 3 \pmod 4$.* Describe the action of $T(p)$ ($p \equiv 3 \pmod 4$) on the Galois orbits, in terms of Frobenius elements.
(3) *Extension to other metaplectic groups.* Apply the same strategy to $A_4'$, $A_5'$, and other metaplectic covers arising in the modular flavour literature.

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